

# Daily Logic Drill #4 — Answers & Investigations

Sheet by David Akinyele-Aje

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

Answers and Methods.

## Section A Rapid Recognition

1. Give a counterexample to: “Every prime number is odd.”

**Answer:**  $n = 2$ .

**Method:** 2 is prime but even, breaking the universal claim with a single instance.

**Investigate further:** 2 is the unique even prime (Day 1, B9) — almost every “all primes are odd”-flavoured false claim in this pillar breaks exactly here.

2. Give a counterexample to: “ $n^2 \geq n$  for every real number  $n$ .”

**Answer:**  $n = 0.5$ .

**Method:**  $n^2 = 0.25$  and  $n = 0.5$ , so  $n^2 < n$ , breaking the claim. (Over the integers this claim is actually true, since  $n^2 - n = n(n - 1) \geq 0$  always — the failure only appears strictly between 0 and 1.)

**Investigate further:** This is a genuinely different claim from its integer version — restricting or widening the domain of a universal claim can flip it from true to false.

3. Give a counterexample to: “If  $n$  is prime, then  $n + 2$  is prime.”

**Answer:**  $n = 7$ .

**Method:** 7 is prime, but  $7 + 2 = 9 = 3^2$  is not prime.

**Investigate further:** This is exactly the twin-prime claim in reverse — most primes do *not* have a prime 2 higher, which is why twin primes are notable rather than universal.

4. Give a counterexample to: “The sum of two primes is always even.”

**Answer:**  $2 + 3 = 5$ .

**Method:** 2 and 3 are both prime, and their sum 5 is odd.

**Investigate further:** Every other pair of odd primes does sum to an even number — 2’s exceptional parity is again the single failure point.

5. Give a counterexample to: “Every multiple of 4 is a multiple of 8.”

**Answer:**  $n = 4$ .

**Method:** 4 is a multiple of 4 but not of 8.

**Investigate further:** This restates Day 1's A9/B6-style fact in explicit counterexample form — the smallest possible instance is usually the cleanest counterexample.

6. Give a counterexample to: “ $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$  for all  $a, b \geq 0$ .”

**Answer:**  $a = b = 1$ .

**Method:**  $\sqrt{1+1} = \sqrt{2} \approx 1.41$ , but  $\sqrt{1} + \sqrt{1} = 2$ ; these are unequal.

**Investigate further:** The correct relationship is  $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$  (with equality only when  $a = 0$  or  $b = 0$ ) — a genuine inequality, not an identity.

7. Give a counterexample to: “ $(x+y)^2 = x^2 + y^2$  for all real  $x, y$ .”

**Answer:**  $x = y = 1$ .

**Method:**  $(1+1)^2 = 4$ , but  $1^2 + 1^2 = 2$ ; these are unequal.

**Investigate further:** The correct expansion is  $(x+y)^2 = x^2 + 2xy + y^2$  — the claim silently drops the cross term  $2xy$ , which vanishes only when  $x = 0$  or  $y = 0$ .

8. Give a counterexample to: “ $n! + 1$  is always prime.”

**Answer:**  $n = 4$ .

**Method:**  $4! + 1 = 24 + 1 = 25 = 5^2$ , which is not prime.

**Investigate further:** This is a famous near-miss pattern:  $n! + 1$  is prime for  $n = 1, 2, 3$ , making the false universal claim tempting before it breaks at  $n = 4$  — Section D revisits this exact claim.

9. Give a counterexample to: “If a number is not prime, it must be even.”

**Answer:**  $n = 9$ .

**Method:**  $9 = 3 \times 3$  is not prime, but 9 is odd.

**Investigate further:** Odd composite numbers (like 9, 15, 21, 25, ...) are the entire reason this claim fails — “not prime” includes far more than “even.”

10. Give a counterexample to: “All odd numbers greater than 1 are prime.”

**Answer:**  $n = 9$ .

**Method:** 9 is odd and greater than 1, but  $9 = 3 \times 3$  is not prime.

**Investigate further:** The same instance as A9 disproves both claims at once — a good counterexample often kills more than one false universal simultaneously.

## Section B Manipulation Drills

1. Give a counterexample to: “ $n^2 + n + 41$  is prime for every positive integer  $n$ .”

**Answer:**  $n = 41$ .

**Method:** At  $n = 41$ :  $n^2 + n + 41 = 41^2 + 41 + 41 = 41(41 + 1) + 41 = 41 \times 42 + 41 = 41 \times 43$ , manifestly divisible by 41 and hence not prime.

**Investigate further:** Substituting  $n = 41$  (matching the constant term) is a structural trick, not luck: whenever  $n \equiv 0 \pmod{41}$ , every term in  $n^2 + n + 41$  is divisible by 41. Section C finds an even smaller failure.

2. Give a counterexample to: “If  $a$  and  $b$  are both irrational, then  $a + b$  is irrational.”

**Answer:**  $a = \sqrt{2}$ ,  $b = -\sqrt{2}$ .

**Method:** Both  $\sqrt{2}$  and  $-\sqrt{2}$  are irrational, but  $a + b = 0$ , which is rational.

**Investigate further:** Deliberately choosing a pair that cancels is a standard counterexample-construction move whenever a claim involves addition.

3. Give a counterexample to: “If  $a$  and  $b$  are both irrational, then  $ab$  is irrational.”

**Answer:**  $a = b = \sqrt{2}$ .

**Method:**  $\sqrt{2}$  is irrational, but  $ab = \sqrt{2} \times \sqrt{2} = 2$ , which is rational.

**Investigate further:** The multiplicative cousin of B2: choosing  $a = b$  (or reciprocal-like pairs) is the standard move when a claim involves a product.

4. Give a counterexample to: “Every quadratic with integer coefficients that has an integer root must have two integer roots.”

**Answer:**  $2x^2 - 3x + 1$ .

**Method:**  $2x^2 - 3x + 1 = (2x - 1)(x - 1)$ , with roots  $x = 1$  (an integer) and  $x = \frac{1}{2}$  (not an integer).

**Investigate further:** The claim is actually true for *monic* quadratics (leading coefficient 1): an integer root forces the other root to be  $(-b - r)$ , also an integer, by Vieta. Non-monic coefficients are exactly what breaks it.

5. Give a counterexample (using  $n = 0$ ) to: “For every integer  $n$ ,  $n/n = 1$ .”

**Answer:**  $n = 0$ .

**Method:**  $0/0$  is undefined, not equal to 1, so the claim fails at  $n = 0$ .

**Investigate further:** Division by zero is the single most common source of “for every  $n$ ” claims quietly failing — always check  $n = 0$  first when a claim involves a fraction with  $n$  in the denominator.

6. Give a counterexample (using  $n = 1$ ) to: “Every positive integer has at least one prime factor.”

**Answer:**  $n = 1$ .

**Method:** 1 has no prime factors at all (its only positive divisor is itself, and 1 is not prime).

**Investigate further:** 1 is neither prime nor composite — it is the standard edge case that breaks “every positive integer has property X (derived from prime factorisation)” claims.

7. Give a counterexample (using  $n = -1$ ) to: “For all integers  $n$ ,  $n^n$  is positive.”

**Answer:**  $n = -1$ .

**Method:**  $(-1)^{-1} = \frac{1}{-1} = -1$ , which is negative, not positive.

**Investigate further:** Negative exponents flip a base into a reciprocal, and a negative base raised to an odd power stays negative — both effects compound at  $n = -1$ .

8. Give a counterexample (using a non-integer) to: “If  $x^2$  is an integer, then  $x$  is an integer.”

**Answer:**  $x = \sqrt{2}$ .

**Method:**  $x^2 = 2$  is an integer, but  $x = \sqrt{2}$  is irrational, hence not an integer.

**Investigate further:** This is exactly the converse-direction failure behind Day 2’s D2 classification (“ $x$  rational” is sufficient but not necessary for “ $x^2$  rational”) — here sharpened to integers specifically.

9. Give a counterexample to: “If  $p$  and  $q$  are distinct primes, then  $p + q$  is composite.”

**Answer:**  $p = 2, q = 3$ .

**Method:** 2 and 3 are distinct primes, but  $2 + 3 = 5$  is itself prime, not composite.

**Investigate further:** Once again 2’s exceptional evenness is the entire failure — for any two distinct *odd* primes, the sum is even and  $\geq 8$ , hence genuinely composite.

10. Give a counterexample to: “For every integer  $n \geq 1$ ,  $2^n - 1$  is prime.”

**Answer:**  $n = 4$ .

**Method:**  $2^4 - 1 = 15 = 3 \times 5$ , which is not prime.

**Investigate further:** Primes of the form  $2^n - 1$  (Mersenne primes) require  $n$  itself to be prime as a necessary — but, per C4, not sufficient — condition;  $n = 4$  fails at the first hurdle since 4 isn’t even prime.

## Section C Substitution & Structure

1. Find the smallest positive integer  $n$  for which  $n^2 + n + 41$  is *not* prime.

**Answer:**  $n = 40$ .

**Method:** At  $n = 40$ :  $40^2 + 40 + 41 = 1600 + 40 + 41 = 1681 = 41^2$ , not prime. This is the smallest failure of Euler’s famous prime-generating polynomial  $n^2 + n + 41$ , which is prime for every  $n = 0, 1, \dots, 39$  before failing here.

**Investigate further:**  $n = 41$  (B1) is an easier-to-justify failure algebraically, but  $n = 40$  is the genuinely smallest one — “a counterexample” and “the smallest counterexample” can demand very different amounts of work.

2. Give a counterexample to: “If  $a \equiv b \pmod{n}$ , then  $a^2 \equiv b^2 \pmod{n^2}$ .”

**Answer:**  $n = 3, a = 1, b = 4$ .

**Method:**  $1 \equiv 4 \pmod{3}$  (both leave remainder 1). But  $a^2 = 1$  and  $b^2 = 16$ , and modulo  $n^2 = 9$ :  $16 \equiv 7 \pmod{9}$ , while  $1 \equiv 1 \pmod{9}$ ; since  $1 \not\equiv 7 \pmod{9}$ ,  $a^2 \not\equiv b^2 \pmod{9}$ .

**Investigate further:** Squaring a congruence modulo  $n$  only guarantees the square modulo  $n$  itself (that part is always safe), never automatically modulo  $n^2$ .

3. Which of the following values of  $n$  is a counterexample to “ $n^2 - n + 11$  is prime for every positive

integer  $n$ ”?

**Answer:** C)  $n = 11$ .

**Method:** At  $n = 11$ :  $n^2 - n + 11 = 11^2 - 11 + 11 = 11^2 = 121$ , since the  $-n + 11$  terms cancel exactly when  $n = 11$ , leaving a bare square.  $121 = 11^2$  is not prime. (Options A, B, D give 11, 31, 101 respectively, all prime — only C fails.)

**Investigate further:** Whenever  $n$  matches the constant term of  $n^2 - n + c$ , the two linear terms cancel and the expression collapses to  $n^2 = c^2$  — always a perfect square, and prime only if  $c = 1$ .

4. Give a counterexample to: “If  $p$  is prime, then  $2^p - 1$  is prime.”

**Answer:**  $p = 11$ .

**Method:** 11 is prime, but  $2^{11} - 1 = 2047 = 23 \times 89$ , which is not prime.

**Investigate further:** This shows primality of  $p$  is necessary but not sufficient for  $2^p - 1$  to be prime (Day 2’s classification language) — exactly the fact Day 4’s D5 builds on.

5. Give a counterexample to: “If  $a^2 \equiv b^2 \pmod{n}$ , then  $a \equiv b \pmod{n}$ .”

**Answer:**  $n = 8, a = 1, b = 3$ .

**Method:**  $a^2 = 1$  and  $b^2 = 9 \equiv 1 \pmod{8}$ , so  $a^2 \equiv b^2 \pmod{8}$ . But  $a \equiv 1$  and  $b \equiv 3 \pmod{8}$ , and  $1 \not\equiv 3 \pmod{8}$ , so  $a \not\equiv b \pmod{8}$ .

**Investigate further:** The correct forced conclusion is only  $a \equiv \pm b \pmod{n}$  (or more generally, that  $n$  divides  $(a - b)(a + b)$ ) — exactly the “lost sign” pattern from Day 1’s B6 (squaring loses sign information), now transplanted into modular arithmetic.

6. Find the smallest positive integer  $n$  for which  $3^n + 2$  is *not* prime.

**Answer:**  $n = 5$ .

**Method:**  $3^1 + 2 = 5$ ,  $3^2 + 2 = 11$ ,  $3^3 + 2 = 29$ ,  $3^4 + 2 = 83$  are all prime.  $3^5 + 2 = 243 + 2 = 245 = 5 \times 49 = 5 \times 7^2$ , which is not prime.

**Investigate further:** Unlike C1’s polynomial, this exponential claim fails much sooner — there’s no general reason to expect “prime for small  $n$ ” patterns to persist, and checking a handful of small cases is often the fastest way to find (or rule out) a break.

7. Give a counterexample to: “If  $x$  and  $y$  are both non-integers, then  $x + y$  is a non-integer.”

**Answer:**  $x = y = 0.5$ .

**Method:** 0.5 is not an integer, but  $x + y = 1$ , which is an integer.

**Investigate further:** Just as with B2’s irrational-cancellation trick, pairing two non-integers with complementary fractional parts is the standard move here.

8. Give a counterexample to: “For all integers  $a, b, c$ : if  $a \mid bc$ , then  $a \mid b$  or  $a \mid c$ .”

**Answer:**  $a = 4, b = 2, c = 2$ .

**Method:**  $bc = 4$ , and  $4 \mid 4$ , so  $a \mid bc$  holds. But  $4 \nmid 2$  and  $4 \nmid 2$ , so neither  $a \mid b$  nor  $a \mid c$  holds.

**Investigate further:** This property (Euclid’s Lemma) genuinely does hold whenever  $a$  is prime —  $a = 4$  breaks it precisely because 4 is composite, foreshadowing why Number Theory’s unique factorisation results all require primality as a hypothesis.

## Section D Challenge

1. Consider the claims: I. “ $n^2 + n + 41$  is prime for every positive integer  $n$ .” II. “ $2^n - 1$  is prime whenever  $n$  is prime.” III. “ $n! + 1$  is prime for every positive integer  $n$ .” Which of I, II, III are true?

**Answer:** H) None of them.

**Method:** I is false (C1: fails at  $n = 40$ ). II is false (C4: fails at  $n = 11$ , a prime, since  $2^{11} - 1 = 2047 = 23 \times 89$ ). III is false (A8: fails at  $n = 4$ , since  $4! + 1 = 25 = 5^2$ ). All three are false.

**Investigate further:** Each of I, II, III is a genuinely famous “looks like it should always work” claim in number theory — this question is a reminder that a pattern holding for the first several small cases is never a proof.

2. Find the smallest positive integer  $n$  for which  $n^2 + n + 1$  is *not* prime.

**Answer:**  $n = 4$ .

**Method:**  $n = 1 : 3$  (prime).  $n = 2 : 7$  (prime).  $n = 3 : 13$  (prime).  $n = 4 : 21 = 3 \times 7$ , not prime.

**Investigate further:** Compare with C1’s  $n^2 + n + 41$ : shrinking the constant term from 41 to 1 shrinks the “run” of consecutive primes from 40 terms down to just 3 — the constant term controls how long the illusion of a pattern survives.

3. Give a counterexample to: “The sum of any three consecutive prime numbers is always even.”

**Answer:**  $3 + 5 + 7 = 15$ .

**Method:** 3, 5, 7 are three consecutive primes, and their sum is 15, which is odd.

**Investigate further:** The sum of three odd numbers is always odd; the claim is only ever true when 2 is among the three primes (e.g.  $2 + 3 + 5 = 10$ ), which happens exactly once in the entire sequence of primes — yet another appearance of 2’s exceptional parity.

4. Give a counterexample to: “If  $n$  is a positive integer such that  $n^2 + 1$  is prime, then  $n$  must be even.”

**Answer:**  $n = 1$ .

**Method:**  $n = 1$  gives  $n^2 + 1 = 2$ , which is prime, but  $n = 1$  is odd.

**Investigate further:**  $n = 1$  is the smallest positive integer, and (as in B6) is frequently the case a hastily-stated claim forgets to check — most other small examples ( $n = 2$ : 5 prime, even  $n$ ;  $n = 4$ : 17 prime, even  $n$ ) do fit the pattern, which is exactly what makes  $n = 1$  easy to miss.

5. Statement  $T$  (you may take this as given/true): “If  $2^n - 1$  is prime, then  $n$  is prime.” (a) Explain why B10 and C4’s counterexamples do *not* contradict  $T$ . (b) State  $T$ ’s converse, and determine whether it is true. (c) Is “ $2^n - 1$  is prime” necessary and sufficient for “ $n$  is prime”?

**Answer:** (a) Neither counterexample has a true hypothesis and false conclusion for  $T$  itself.

(b) Converse: “If  $n$  is prime, then  $2^n - 1$  is prime.” False (disproved by C4,  $n = 11$ ). (c) Sufficient but not necessary.

**Method:** (a) B10’s  $n = 4$  has  $2^4 - 1 = 15$  composite, so  $T$ ’s hypothesis (“ $2^n - 1$  is prime”) is false there —  $T$  is not violated by a false hypothesis. C4’s  $n = 11$  has  $2^{11} - 1 = 2047$  composite too, so again  $T$ ’s hypothesis is false, not violated. Both examples in fact disprove the *converse* of  $T$  (a prime  $n$  not forcing  $2^n - 1$  prime), not  $T$  itself. (b) The converse swaps hypothesis and conclusion: “if  $n$  prime then  $2^n - 1$  prime.” C4 is exactly a counterexample to this converse:  $n = 11$  is prime, but  $2^{11} - 1 = 2047$  is not. (c) Since  $T$  (sufficiency direction,  $2^n - 1$  prime  $\Rightarrow n$  prime) is true but  $T$ ’s converse (necessity direction) is false, “ $2^n - 1$  prime” is sufficient but not necessary for “ $n$  prime.”

**Investigate further:** This question is the whole pillar in miniature: distinguishing a statement from its converse (Day 1) resolves directly into a necessary/sufficient classification (Day 2), using two counterexamples (Day 4) found earlier on this very sheet.